

Shawkat Ali

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SUMMARY

Ph.D. in Control Science and Engineering, specializing in signal processing, electrical machine condition monitoring, and induction motor fault diagnosis. Experienced in academic research, teaching, embedded systems, telecommunications, and machine learning applications. Seeking a postdoctoral, industry R&D, or teaching position where I can apply expertise in intelligent diagnostics and electrical machine systems.

EDUCATION

Ph.D. Control Science and Engineering: Jiangsu University, Zhenjiang, P.R. China.

September 2020 – September 2026.

Dissertation: *Robust demodulation strategies for incipient broken rotor bar faults in induction motors.*

Honors: CSC Scholarship (2023-2026).

M.Sc. in Computer Systems Engineering: University of Engineering & Technology, Peshawar, Pakistan.

February 2016 – November 2018.

Thesis: *Consensus control of multi-agent systems.*

Honors: HEC Pakistan-funded project.

B.Sc. in Computer Systems Engineering: University of Engineering & Technology, Peshawar, Pakistan.

September 2010 – June 2014.

Thesis: *OBD2, Android, and Open ERP-Based Vehicles 3M System.*

Honors: HEC Pakistan-funded project.

PROFESSIONAL & TEACHING EXPERIENCE

Adjunct Faculty (Online): Davis University, September 2026 – Present.

- Teach online course(s) remotely, delivering lectures, course materials, and assessments to a distributed student cohort.

Teaching Assistant: Jiangsu University, Zhenjiang, P.R. China, September 2023 - March 2024.

- Assisted with instruction and grading for a graduate course while pursuing Ph.D. research.
- Supported students with research-related tasks.

DOC Engineer: Global Service Center, Huawei Technologies, Pakistan, August 2022 - March 2023.

- Supported the Network & IT System Integration shared-service delivery department for multi-country telecom operations.
- Led quality assurance control for the PHP GLOBE FBB project, coordinating cross-team documentation and process compliance to reduce delivery errors.

Visiting Lecturer: University of Swat, Pakistan, September 2015 - August 2022.

- Taught core undergraduate courses in computer engineering (7 years), including subjects spanning embedded systems, programming, and digital logic.
- Designed course materials, examinations, and assessment rubrics; supervised student final-year projects.

Research Assistant: UET, Peshawar, Pakistan, August 2014 - August 2015.

- Designed hardware and algorithms for automatic OBD-II protocol implementation, contributing to a government-funded project.
- Co-developed a real-time vehicle monitoring and management system, later published as a book.

PUBLICATIONS

Journal Articles

- **Ali, S.**, Song, X. and Zhao, W. (2026), A Novel Cost-Effective Fault Diagnosis Method for Incipient Detection of Broken Rotor Bar in Induction Motor. IEEJ Trans Elec Electron Eng. https://doi.org/10.1002/tee.70414.
- **S. Ali**, X. Song, W. Zhao and H. Fan, "Orthogonal axis rectification and Goertzel analysis for reliable detection of broken rotor bar faults in induction motor," in Chinese Journal of Electrical Engineering, doi: 10.23919/CJEE.2026.000033.
- **S. Ali**, X. Song and W. Zhao, "A dual-signature rotor fault diagnosis method using envelope modulation and slip confirmation," in Chinese Journal of Electrical Engineering, doi: 10.23919/CJEE.2026.000107.

Conference Paper

- **S. Ali**, S. Ahmed, and S. N. Khan Marwat, "A practical approach to consensus based control of multi-agent systems," 2018 International Symposium on Recent Advances in Electrical Engineering, 2018, pp. 1-5.

Book

- Haq. A.U., **S. Ali**, Rehman: OBDII, Android, and Open ERP-Based Vehicles 3M System (2015).

RESEARCH EXPERIENCE & PROJECTS

Robust demodulation strategies for incipient broken rotor bar faults in induction motors – Ph.D. Research, Jiangsu University, 2023 – 2026.

- Developed high-resolution, cost-effective signal-processing methods for early diagnosis of broken rotor bar faults in induction motors, resulting in three journal papers.
- Designed and validated diagnostic algorithms combining envelope modulation, Goertzel analysis, and orthogonal axis rectification techniques.

Consensus control of multi-agent systems – M.Sc. Research, UET Peshawar, 2016 – 2018

- Investigated distributed consensus strategies to address power mismatch problems in smart grid systems.
- Published findings at the 2018 International Symposium on Recent Advances in Electrical Engineering.

Earlier academic projects

- **Performance Analysis of Classifiers on Brain Tumor Dataset** – Evaluated multiple classifiers, including KNN, Bayesian methods, neural networks, and K-means-based approaches, on MRI datasets.

- **Comparative Study of HEVC and H.264 Codecs** – Analyzed codec performance using metrics such as PSNR and computational efficiency.
- **Speed Analysis of Next-Gen Alignment Tools** – Benchmarked bioinformatics alignment tools including Bowtie2, BWA, SOAP, Subread, and TopHat.
- **OBDII, Android, and OpenERP-Based Vehicles' 3M System** – Developed a real-time vehicle monitoring and management system under a government-funded project in Pakistan.

RESEARCH INTEREST

- Motor fault diagnosis, condition monitoring of electrical machines, signal processing, machine learning for diagnostics, and intelligent industrial monitoring.

HONORS AND AWARDS

- Excellent Project Award at the Discover Jiangsu International Summer Program, Jiangsu University, 2024.
- CSC (China Scholarship Council) Scholarship, P.R. China (2023-2026).
- Travel Grant / Research Assistantship Support, Office of Graduate Studies, UET Peshawar, Pakistan, 2018.
- Final Year Project distinction under an HEC Pakistan-funded project, 2014.

PROFESSIONAL SERVICE

- Peer Reviewer: IEEE Access, IEEE TIM, IEEE JERM, ERX, Microsoft CMT.
- President, Computer Society, UET Chapter, UET Peshawar, Pakistan (2013-2014).
- President, Swat Student Society, UET Peshawar, Pakistan (2013-2014).
- Convocation Co-Organizer, UET Peshawar, Pakistan, 2013.

LANGUAGES

English (Fluent), Urdu/Hindi (Native), Pashto (Native), Arabic (Basic), Chinese (HSK-3).

COMPUTER SKILLS

- **Programming languages:** Assembly, C, C++, C#, Python, Verilog HDL, PHP, HTML
- **Software and Engineering Tools:** MATLAB, LaTeX, TensorFlow, PyTorch, Cisco Packet Tracer, NS-2, Proteus, AutoCAD, Workbench, Modelsim, Keil uVision, Arduino, Unity 3D, WordPress.
- **Platforms:** Linux, Windows.

CERTIFICATIONS

- CCNA Cisco Certified Network Associate 2017.
- Intern, National ICT R&D Fund, Ministry of IT, 2014.
- Certificate of appreciation, UET Peshawar convocation 2013.
- Internee Training Certificate, National Center for Physics, Islamabad, 06/2012-08/2012.
- Special training course (MS Office) Skill development council 06/2007-09/2007.

OTHER ACTIVITIES

Football, Badminton, Table Tennis, Hiking, and Planting.

REFERENCES

1. **Wenxiang Zhao**

   Professor (Ph.D. Supervisor), School of Electrical and Information Engineering, Jiangsu University, Zhenjiang 212013, P.R. China. E-mail: zwx@ujs.edu.cn.

2. **Xiangjin Song**

   Associate Professor (Ph.D. Co-Supervisor), School of Electrical and Information Engineering, Jiangsu University, Zhenjiang 212013, P.R. China. E-mail: songxiangjin@ujs.edu.cn.

3. **Salman Ahmed**

   Associate Professor (Master Supervisor), Faculty of Computer Science & Artificial Intelligence, GIKI, Pakistan. E-mail: sahmed@giki.edu.pk.